DATA Midterm Project

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Set Up

```r
library(ggplot2)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.2.1
## v purrr     1.2.0     v tidyr     1.3.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(dplyr)
```

```r
# reproducibility
set.seed(123)

# Load dataset
# Download dataset from Kaggle if not already present
if (!file.exists("insurance.csv")) {
  url <- "https://raw.githubusercontent.com/stedy/Machine-Learning-with-R-datasets/master/insurance.csv
  download.file(url, destfile = "insurance.csv", method = "curl")}
insurance <- read.csv("insurance.csv")

# Preview the dataset
head(insurance)
```

```
##   age    sex    bmi children smoker    region   charges
## 1  19 female 27.900        0    yes southwest 16884.924
## 2  18   male 33.770        1     no southeast  1725.552
## 3  28   male 33.000        3     no southeast  4449.462
## 4  33   male 22.705        0     no northwest 21984.471
## 5  32   male 28.880        0     no northwest  3866.855
## 6  31 female 25.740        0     no southeast  3756.622
```

```r
str(insurance)
```

```
## 'data.frame':    1338 obs. of  7 variables:
##  $ age     : int  19 18 28 33 32 31 46 37 37 60 ...
##  $ sex     : chr  "female" "male" "male" "male" ...
##  $ bmi     : num  27.9 33.8 33 22.7 28.9 ...
##  $ children: int  0 1 3 0 0 0 1 3 2 0 ...
##  $ smoker  : chr  "yes" "no" "no" "no" ...
##  $ region  : chr  "southwest" "southeast" "southeast" "northwest" ...
##  $ charges : num  16885 1726 4449 21984 3867 ...
```

```r
summary(insurance)
```

```
##       age            sex                 bmi           children    
##  Min.   :18.00   Length:1338        Min.   :15.96   Min.   :0.000  
##  1st Qu.:27.00   Class :character   1st Qu.:26.30   1st Qu.:0.000  
##  Median :39.00   Mode  :character   Median :30.40   Median :1.000  
##  Mean   :39.21                      Mean   :30.66   Mean   :1.095  
##  3rd Qu.:51.00                      3rd Qu.:34.69   3rd Qu.:2.000  
##  Max.   :64.00                      Max.   :53.13   Max.   :5.000  
##     smoker             region             charges     
##  Length:1338        Length:1338        Min.   : 1122  
##  Class :character   Class :character   1st Qu.: 4740  
##  Mode  :character   Mode  :character   Median : 9382  
##                                        Mean   :13270  
##                                        3rd Qu.:16640  
##                                        Max.   :63770  
```

Data contains 1338 observations and 7 variables. The numerical variables are age, bmi, children, and charges. The categorical variables are sex, smoker, region. This dataset provides demographic and lifestyle information for individuals corresponding with their medical insurance cost billed in the US. All variables are loaded correctly and there are no missing values.

Data Cleaning

```r
insurance <- insurance %>%
  mutate(
    sex = factor(sex),
    smoker = factor(smoker),
    region = factor(region))

str(insurance)
```

```
## 'data.frame':    1338 obs. of  7 variables:
##  $ age     : int  19 18 28 33 32 31 46 37 37 60 ...
##  $ sex     : Factor w/ 2 levels "female","male": 1 2 2 2 2 1 1 1 2 1 ...
##  $ bmi     : num  27.9 33.8 33 22.7 28.9 ...
##  $ children: int  0 1 3 0 0 0 1 3 2 0 ...
##  $ smoker  : Factor w/ 2 levels "no","yes": 2 1 1 1 1 1 1 1 1 1 ...
##  $ region  : Factor w/ 4 levels "northeast","northwest",..: 4 3 3 2 2 3 3 2 1 2 ...
##  $ charges : num  16885 1726 4449 21984 3867 ...
```

I converted the character columns (sex, smoker, region) into factors to make sure they're treated as categorical predictors. It allows R to create dummy variables automatically when fitting regression models. Each factor level (ex. "male", "female") will have its own coefficient.

Exploratory Data Analysis

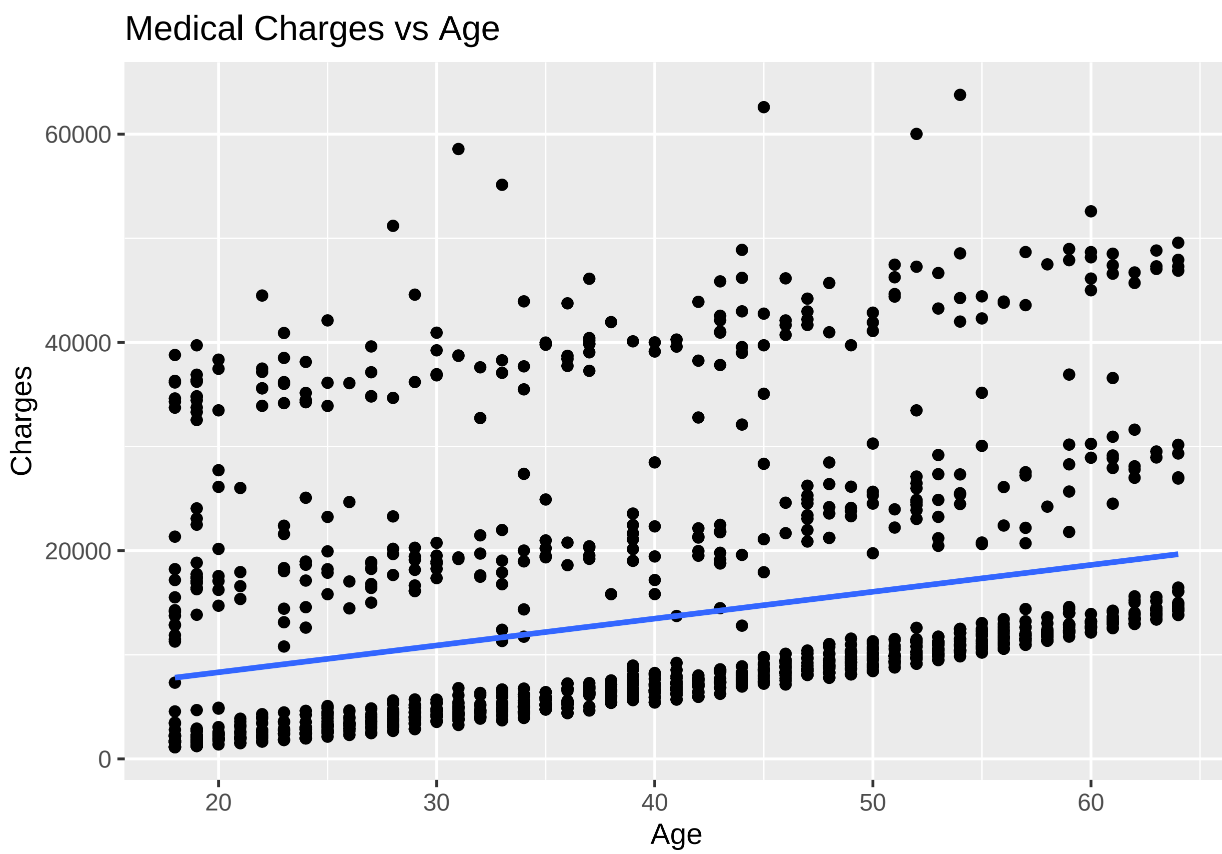
```r
summary(insurance)
```

```
##       age            sex           bmi           children     smoker
##  Min.   :18.00   female:662   Min.   :15.96   Min.   :0.000   no :1064
##  1st Qu.:27.00   male  :676   1st Qu.:26.30   1st Qu.:0.000   yes: 274
##  Median :39.00                Median :30.40   Median :1.000
##  Mean   :39.21                Mean   :30.66   Mean   :1.095
##  3rd Qu.:51.00                3rd Qu.:34.69   3rd Qu.:2.000
##  Max.   :64.00                Max.   :53.13   Max.   :5.000
##        region       charges
##  northeast:324   Min.   : 1122
##  northwest:325   1st Qu.: 4740
##  southeast:364   Median : 9382
##  southwest:325   Mean   :13270
##                  3rd Qu.:16640
##                  Max.   :63770
```

Mean age is 39, mean BMI is 30.7, there are 274 smokers and 1064 non-smokers, and the average medical charge is $13,270. There is a large range that goes from $1,122 to $63,770. Since the medical costs vary so widely, there could be a skew from a few individuals with extremely high expenses. They tested a similar number of males and females and a similar number of people from each region, so the data seems well spread out.

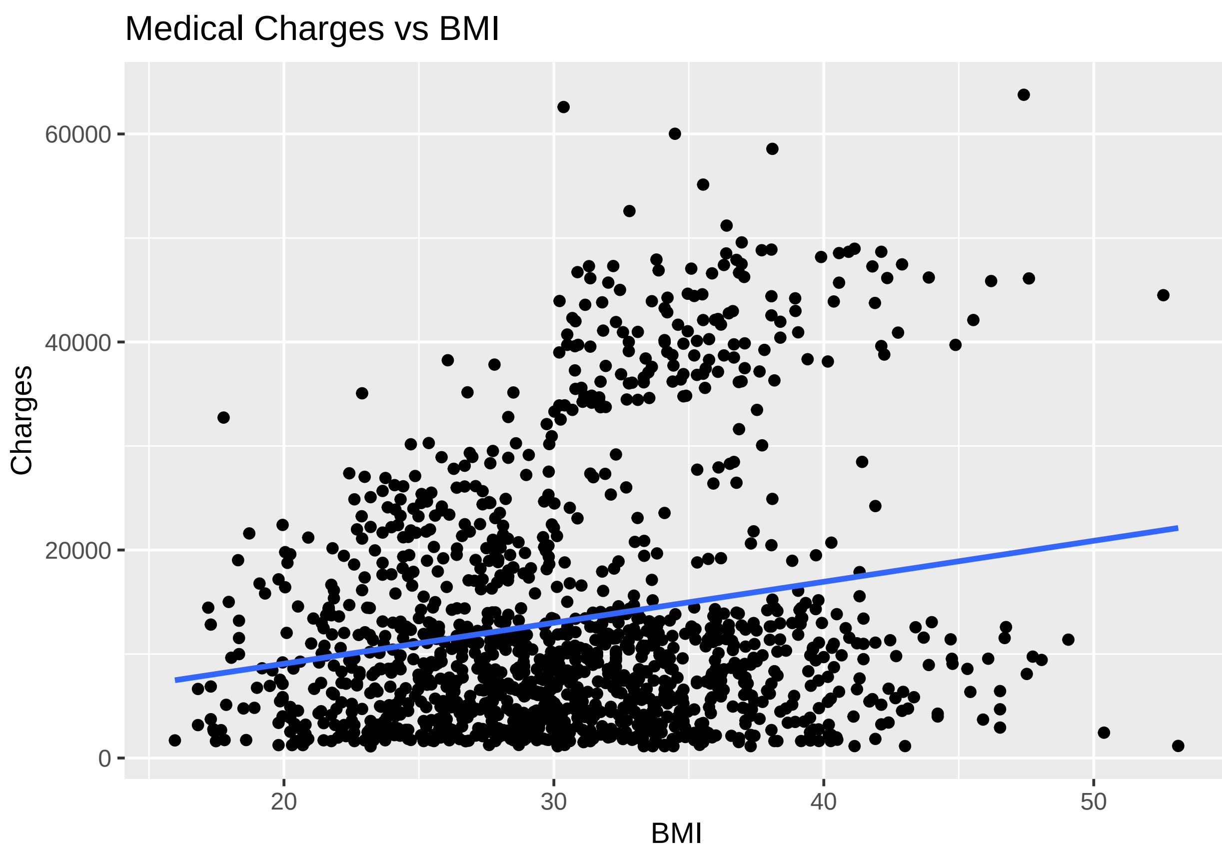
```r
# Looking at Relationships Visually
# Age vs. Charges
ggplot(insurance, aes(x = age, y = charges)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Medical Charges vs Age", x = "Age", y = "Charges")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



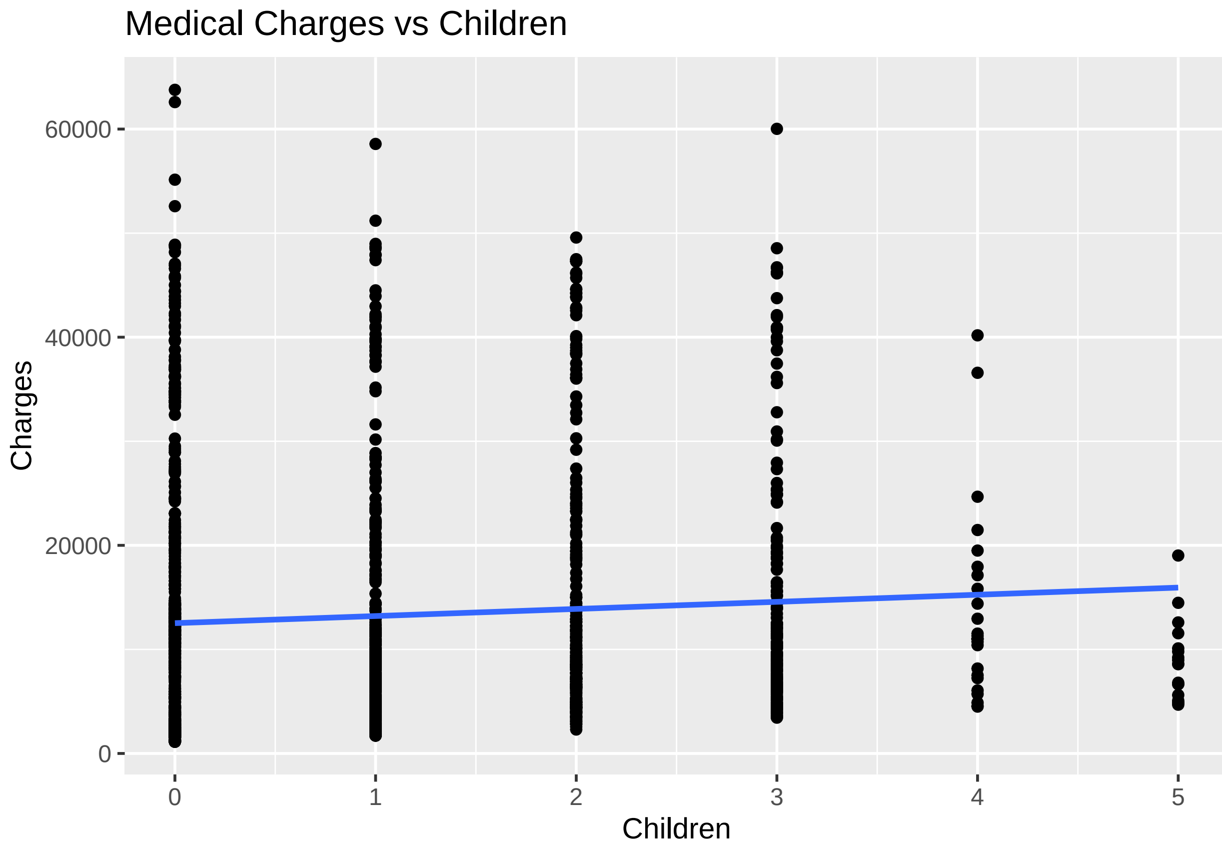

```
# BMI vs. Charges
ggplot(insurance, aes(x = bmi, y = charges)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Medical Charges vs BMI", x = "BMI", y = "Charges")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

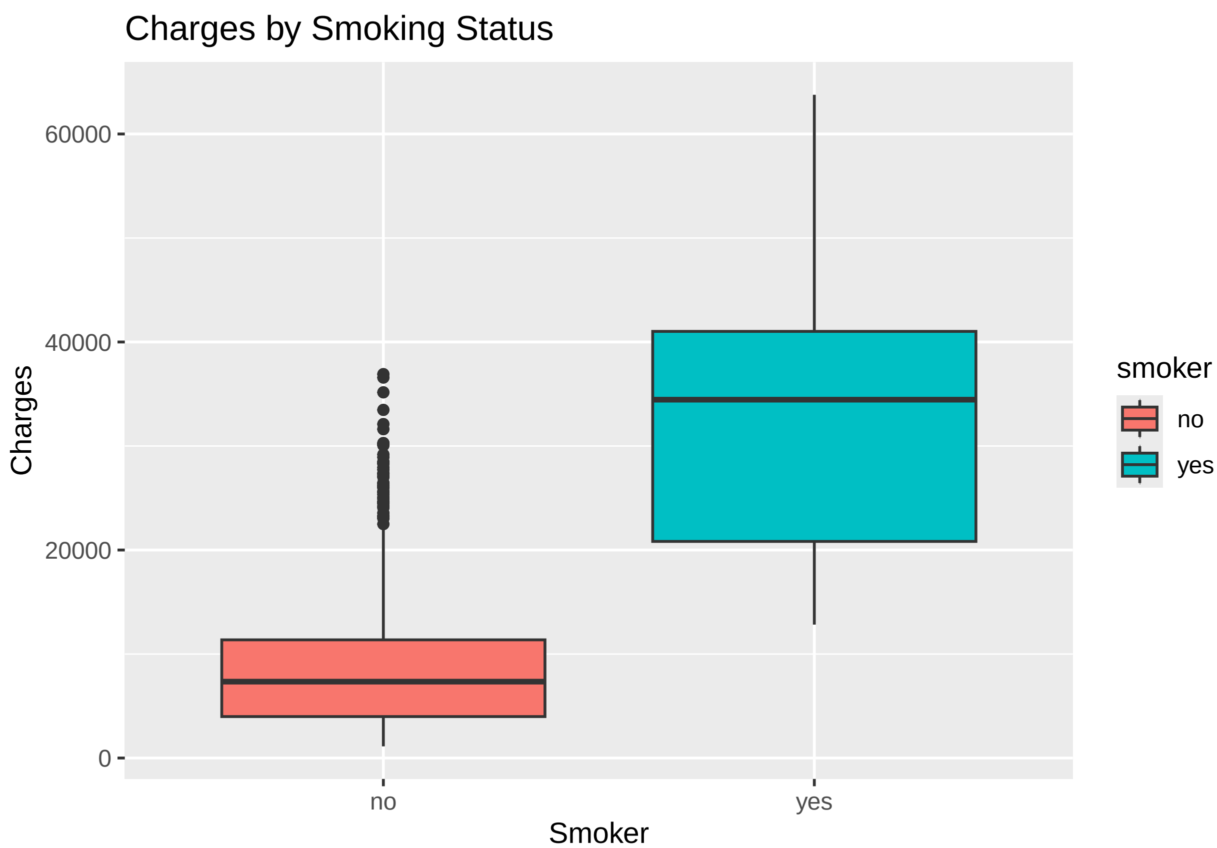



```r
# Children vs. Charges
ggplot(insurance, aes(x = children, y = charges)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Medical Charges vs Children", x = "Children", y = "Charges")
```

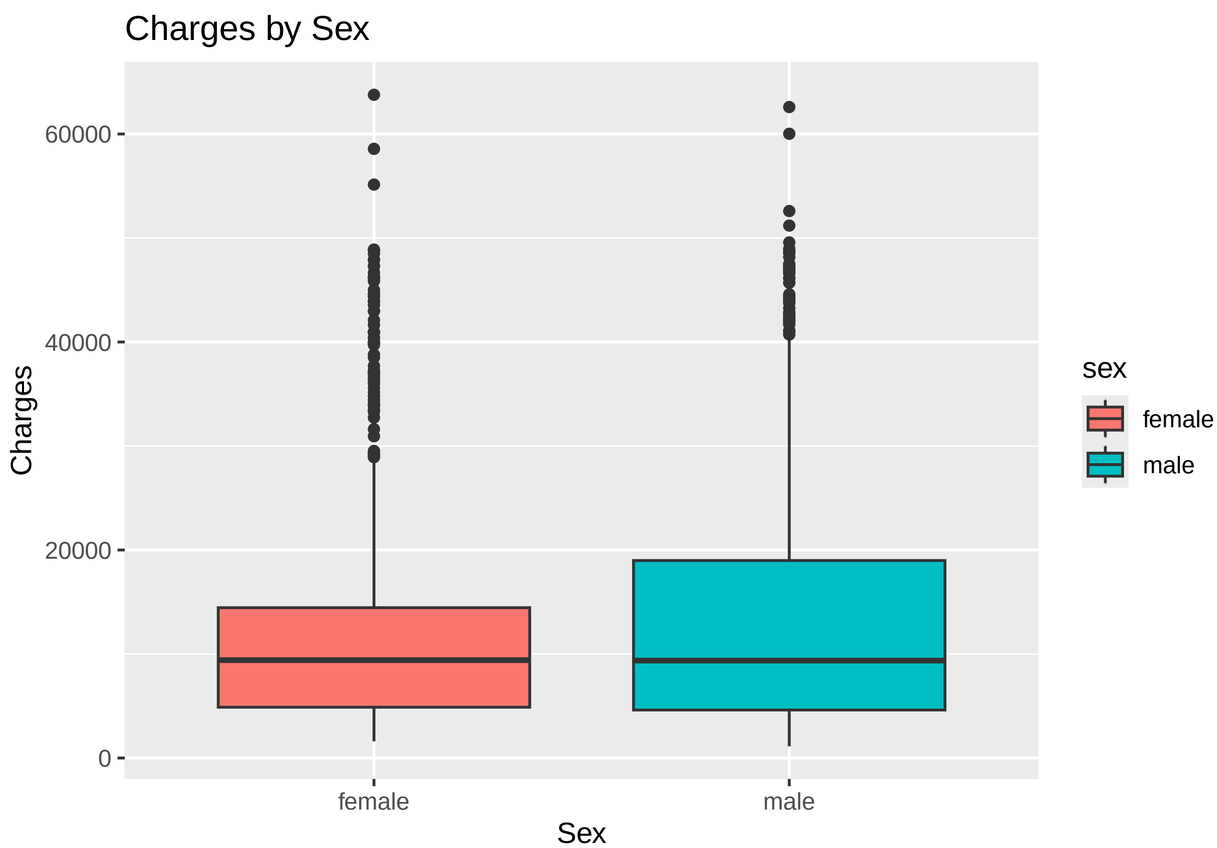
```
## `geom_smooth()` using formula = 'y ~ x'
```



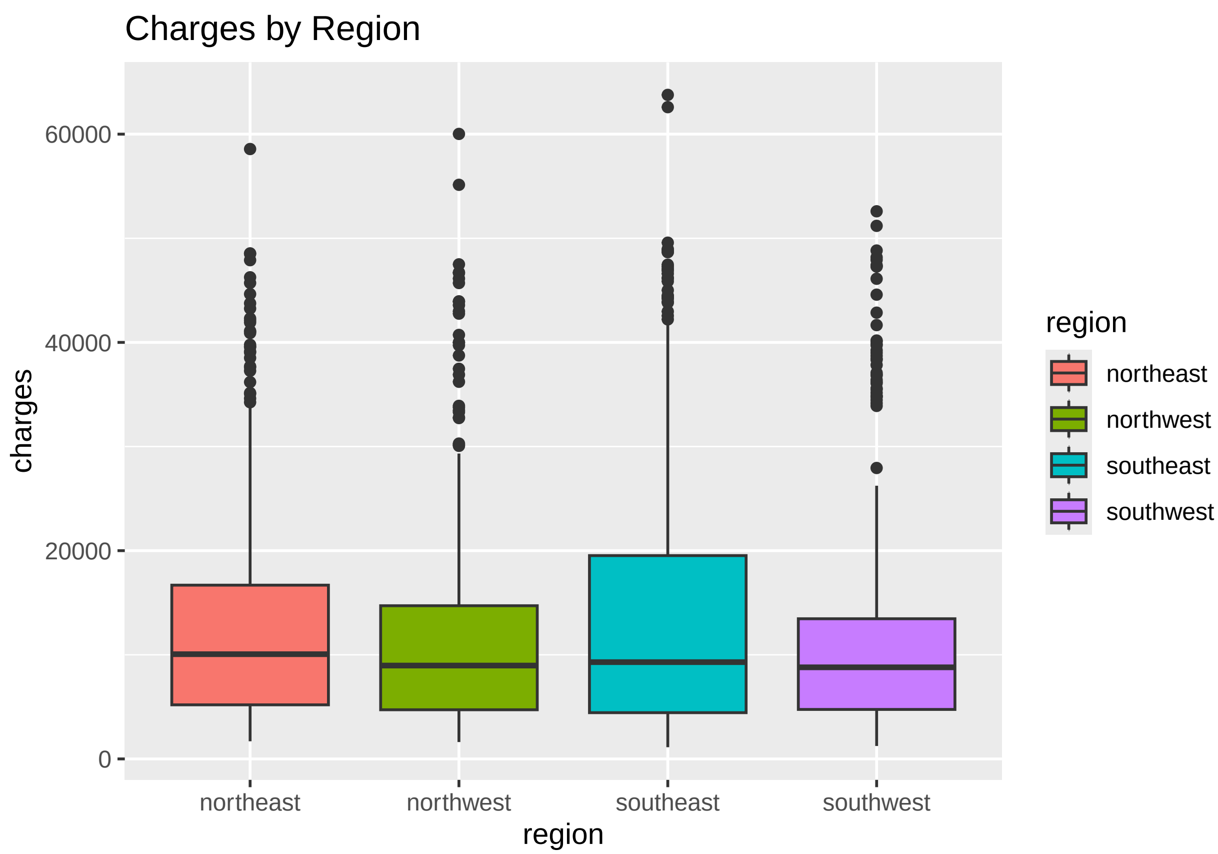

```r
# Smoker vs. Charges
ggplot(insurance, aes(x = smoker, y = charges, fill = smoker)) +
  geom_boxplot() +
  labs(title = "Charges by Smoking Status", x = "Smoker", y = "Charges")
```




```r
# Sex vs. Charges
ggplot(insurance, aes(x = sex, y = charges, fill = sex)) +
  geom_boxplot() +
  labs(title = "Charges by Sex", x = "Sex", y = "Charges")
```




```r
# Region vs. Charges
ggplot(insurance, aes(x = region, y = charges, fill = region)) +
  geom_boxplot() +
  labs(title = "Charges by Region")
```



Age vs Charges: scatterplot shows a positive relationship (as age increases, the insurance charges rise). The fitted line shows a roughly linear trend. Variability in charges increases with age since older individuals show both very low and very high costs. The increasing spread of points suggests mild heteroscedasticity.

BMI vs Charges: positive relationship (higher BMI corresponds to higher charges). Some high BMI individuals (outliers) have extremely high charges. The pattern suggests that medical expenses rise non-linearly after a certain BMI, although a linear term still captures the general relationship. The increase in charges is sharper beyond a BMI of ~30.

Children vs. Charges: The trend line implies that as the number of children increases, medical charges tend to rise slightly. However, the spread of data points looks quite wide—suggesting low explanatory power.

Smoker vs. Charges: smokers have dramatically higher charges than non-smokers. This variable will likely be the strongest predictor in the model. Non-smokers cluster mostly below $15,000, while smokers range up to $60,000 or more. The visual also hints at possible interaction with BMI.

Sex vs. Charges: both distributions are right-skewed, the wider spread for males might suggest unequal variances. The median medical charges are slightly higher for males than for females, suggesting that on average, males incur more medical expenses.

Region vs. Charges: regional differences are minor, the distributions heavily overlap, region probably only has a small effect on cost. Including region allows control for unobserved cost differences but is not expected to meaningfully improve model fit.

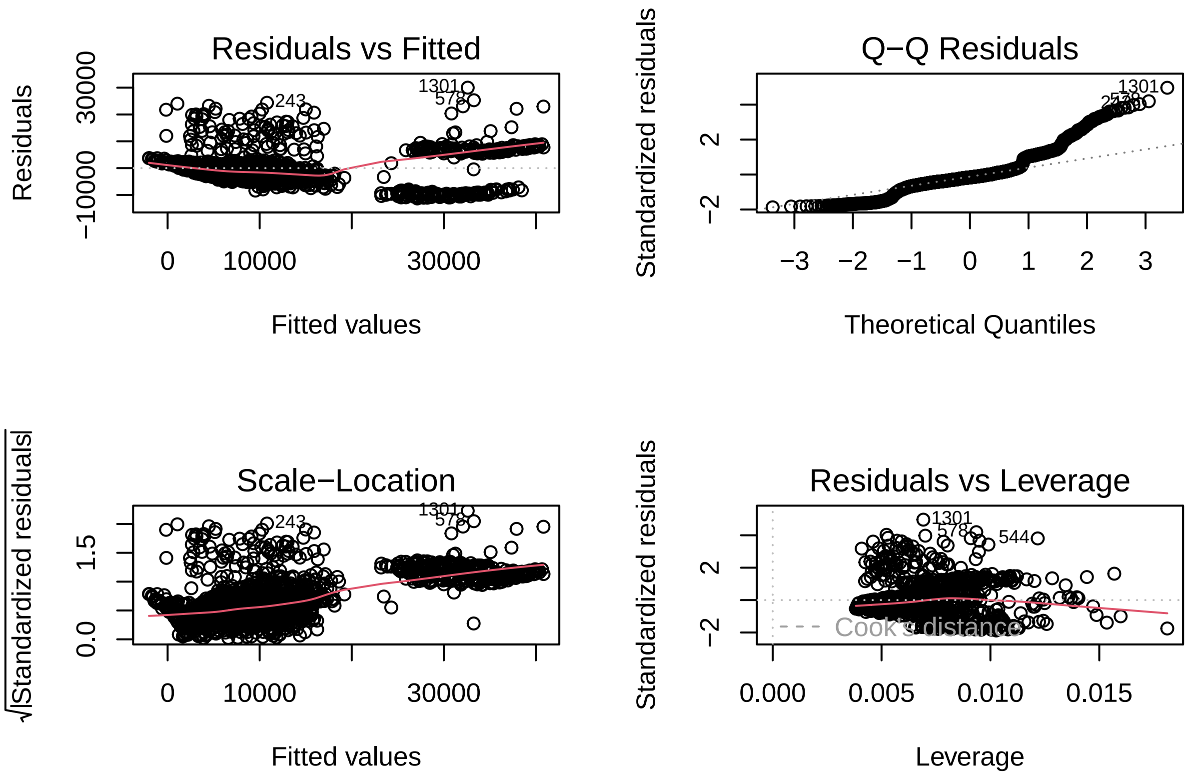
Multiple Linear Regression Model

```r
full_model <- lm(charges ~ age + sex + bmi + children + smoker + region, data = insurance)
summary(full_model)
```

```
##
## Call:
## lm(formula = charges ~ age + sex + bmi + children + smoker +
##     region, data = insurance)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11304.9  -2848.1   -982.1   1393.9  29992.8
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -11938.5      987.8 -12.086  < 2e-16 ***
## age                256.9       11.9  21.587  < 2e-16 ***
## sexmale           -131.3      332.9  -0.394 0.693348
## bmi                339.2       28.6  11.860  < 2e-16 ***
## children           475.5      137.8   3.451 0.000577 ***
## smokeryes        23848.5      413.1  57.723  < 2e-16 ***
## regionnorthwest   -353.0      476.3  -0.741 0.458769
## regionsoutheast  -1035.0      478.7  -2.162 0.030782 *
## regionsouthwest   -960.0      477.9  -2.009 0.044765 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6062 on 1329 degrees of freedom
## Multiple R-squared:  0.7509, Adjusted R-squared:  0.7494
## F-statistic: 500.8 on 8 and 1329 DF,  p-value: < 2.2e-16
```

The residuals range from –11,300 to +29,900 with the mean residual being -982, suggesting a slight left skew in residuals. The interquartile range (IQR) spans from –2,848 to +1,394, showing asymmetry and potential heteroscedasticity. The large spread, especially the maximum positive residual, shows that a few individuals have extremely high charges that the model underpredicts. The strongest predictors are smoker, age, and bmi while sex and most regions show to be non-significant. The effects of smoking increases charges by tens of thousands, compared to hundreds for other predictors. The RSE is relatively high but reasonable given the wide range of charges. The R squared says that the model explains about 75% of the total variations in charges, which is a strong fit. The adjusted R squared is slightly lower, but doesn't really show overfitting. The F-stat shows that the overall model is highly significant, so at least one predictor is strongly related to the outcome. The model is effective in explaining variation in insurance costs.

```r
# Checking model assumptions
par(mfrow = c(2,2))
plot(full_model)
```




```
par(mfrow = c(1,1))
```

Residuals vs Fitted: slight curve around the red line, suggesting mild non-linearity. This violates the linearity assumption. Q-Q Plot: Most points lie on the line, but there's deviation at the tails. Residuals are approximately normal, but heavy tails suggest outliers or skewness. Scale-Location Plot: Mild heteroscedasticity, variance of residuals isn't constant. Residuals vs Leverage: Observations labeled "5440" and "5540" have high leverage and large residuals.

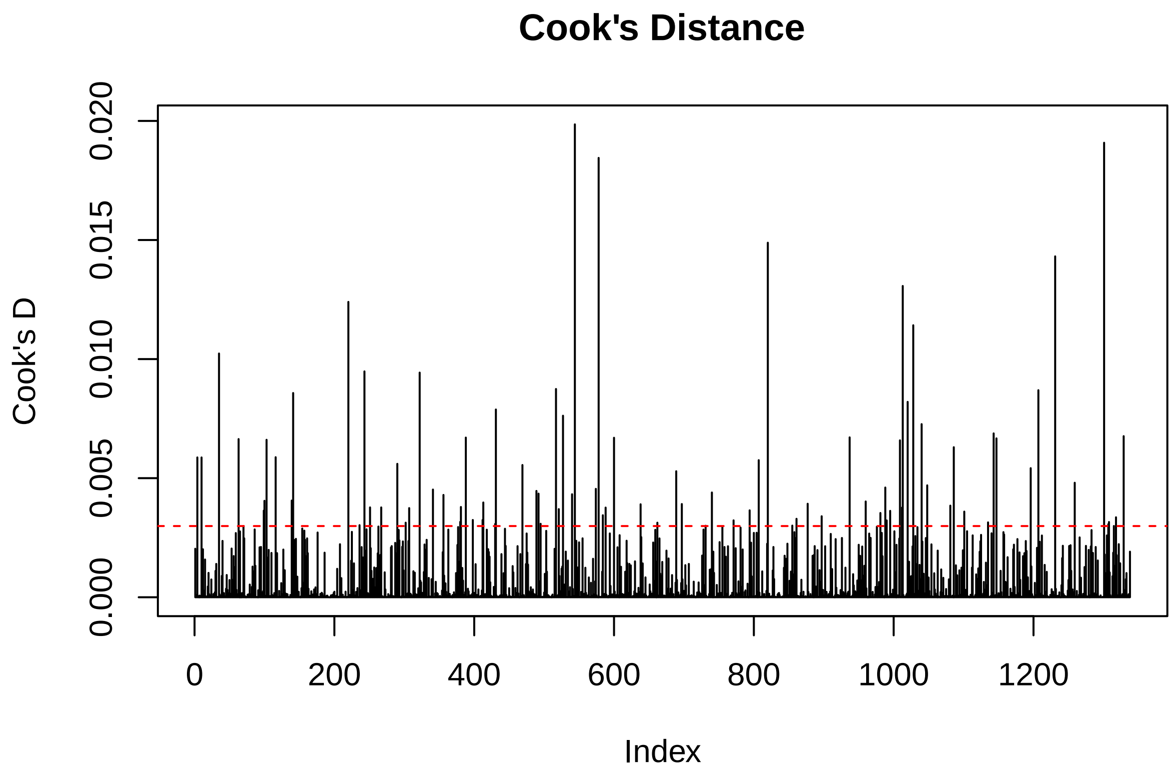
Identify Outliers & Influential Points

```
stud_res <- rstudent(full_model)

# flagging points whose studentized residual exceeds +/- 3 for outliers
which(abs(stud_res) > 3)
```

```
##    4   35  103  141  220  243  388  431  469  517  527  544  578  600  689  820
##    4   35  103  141  220  243  388  431  469  517  527  544  578  600  689  820
##  937  988 1009 1013 1020 1028 1040 1143 1207 1231 1301 1329
##  937  988 1009 1013 1020 1028 1040 1143 1207 1231 1301 1329
```

```
# Cooks Distance
CooksDist <- cooks.distance(full_model)
plot(CooksDist, type = "h", main = "Cook's Distance", ylab = "Cook's D")
abline(h= 4/length(CooksDist), col = "red", lty= 2)
```




```r
#Influential points using Cooks Distance
influential_pt <- which(CooksDist > (4/length(CooksDist)) | abs(stud_res) > 3)
influential_pt
```

```
##    4   10   35   63   70   99  100  103  116  139  141  220  236  243  251  267
##    4   10   35   63   70   99  100  103  116  139  141  220  236  243  251  267
##  290  302  307  322  341  356  380  381  388  398  412  413  430  431  469  489
##  290  302  307  322  341  356  380  381  388  398  412  413  430  431  469  489
##  492  495  517  521  527  540  544  574  578  584  588  600  638  662  689  697
##  492  495  517  521  527  540  544  574  578  584  588  600  638  662  689  697
##  731  740  771  794  807  820  855  861  877  897  937  960  981  988  990  995
##  731  740  771  794  807  820  855  861  877  897  937  960  981  988  990  995
## 1009 1012 1013 1020 1028 1040 1048 1081 1086 1101 1135 1143 1147 1196 1207 1231
## 1009 1012 1013 1020 1028 1040 1048 1081 1086 1101 1135 1143 1147 1196 1207 1231
## 1259 1301 1307 1308 1315 1318 1329
## 1259 1301 1307 1308 1315 1318 1329
```

Flagging observations where |rstudent| > 3, which is a solid threshold for identifying outliers in regression diagnostics. 27 potential outliers with the Cook's distance plot flagging some influential observations (high leverage, large residual). Some points may be both outliers and influential (e.g., 544, 689, 988), which makes them especially important to investigate. 91 observations that are high leverage and influence and/or extreme residuals.

Comparing model without influential points

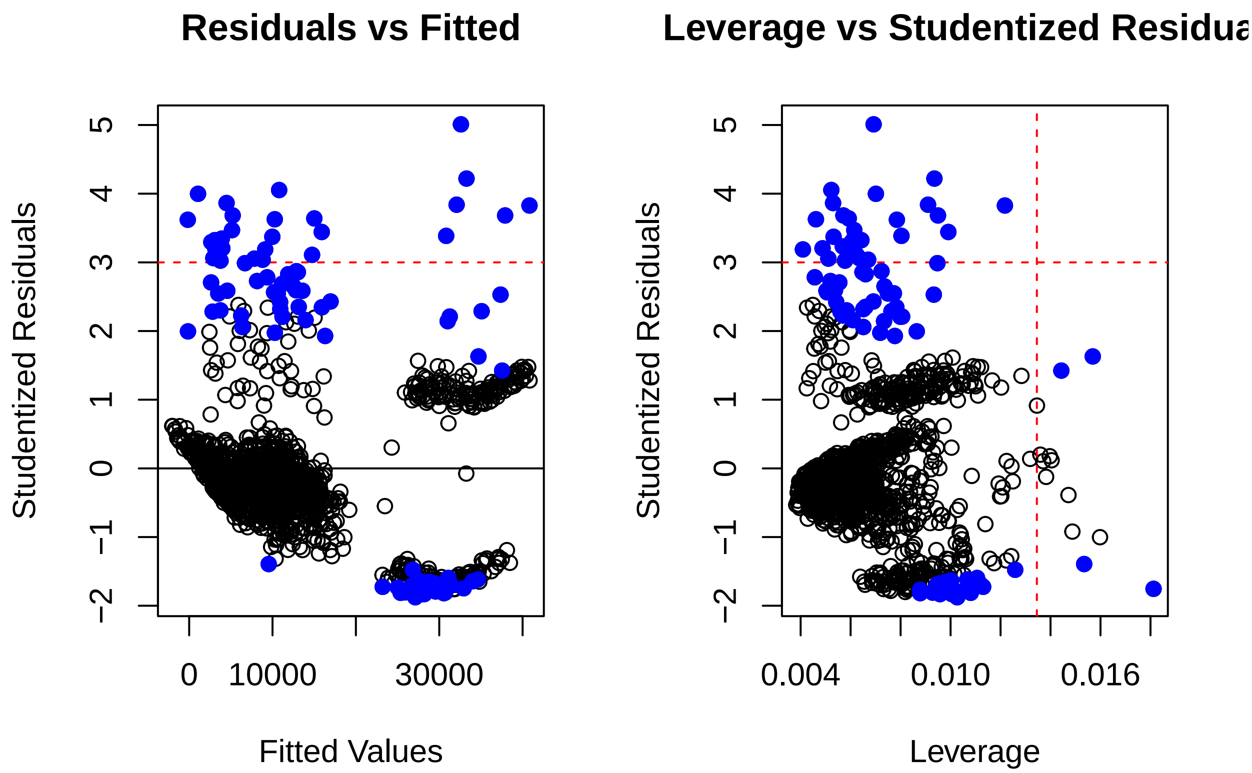
```r
clean_model <- lm(charges ~ age + bmi + children + smoker + region,
                  data = insurance[-influential_pt, ])
summary(clean_model)
```

```
##
## Call:
## lm(formula = charges ~ age + bmi + children + smoker + region,
##     data = insurance[-influential_pt, ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11554.7  -1747.8   -269.4   1567.3  14914.3
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -11054.099    766.923 -14.414  < 2e-16 ***
## age                252.742      9.138  27.658  < 2e-16 ***
## bmi                289.242     22.589  12.804  < 2e-16 ***
## children           518.651    106.362   4.876 1.22e-06 ***
## smokeryes        24795.885    324.361  76.445  < 2e-16 ***
## regionnorthwest   -913.147    366.717  -2.490  0.01290 *
## regionsoutheast  -1054.561    368.845  -2.859  0.00432 **
## regionsouthwest  -1032.353    365.811  -2.822  0.00485 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4496 on 1243 degrees of freedom
## Multiple R-squared:  0.8483, Adjusted R-squared:  0.8475
## F-statistic: 993.1 on 7 and 1243 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(1, 2))

# Plot Residuals vs Fitted
plot(fitted(full_model), stud_res,
     main = "Residuals vs Fitted",
     xlab = "Fitted Values", ylab = "Studentized Residuals")
abline(h = c(-3, 0, 3), col = c("red", "black", "red"), lty = c(2, 1, 2))
points(fitted(full_model)[influential_pt], stud_res[influential_pt],
       col = "blue", pch = 19)

# Plot Leverage vs Studentized Residuals
plot(hatvalues(full_model), stud_res,
     main = "Leverage vs Studentized Residuals",
     xlab = "Leverage", ylab = "Studentized Residuals")
abline(h = c(-3, 3), col = "red", lty = 2)
abline(v = 2 * length(coef(full_model)) / nrow(insurance),
       col = "red", lty = 2)
points(hatvalues(full_model)[influential_pt], stud_res[influential_pt],
       col = "blue", pch = 19)
```




```r
par(mfrow = c(1, 1))
```

Residuals vs Fitted plot: Blue points above the red dashed line at y = 3 are clear outliers. The spread of residuals across fitted values shows mild curvature, suggesting some non-linearity remains. This visualization confirms that flagged points are not randomly scattered, they cluster at higher fitted values, which may reflect high-cost cases (e.g., smokers with high BMI).

Leverage vs Studentized Residuals: Blue points beyond both the horizontal (±3) and vertical lines are high-leverage outliers, they exert disproportionate influence on the regression line.

Comparing Models

```r
step_model <- step(clean_model, direction = "both")
```

```
## Start:  AIC=21051.95
## charges ~ age + bmi + children + smoker + region
##
##            Df  Sum of Sq        RSS   AIC
## <none>                   2.5122e+10 21052
## - region    3 2.2659e+08 2.5348e+10 21057
## - children  1 4.8057e+08 2.5602e+10 21074
## - bmi       1 3.3136e+09 2.8435e+10 21205
## - age       1 1.5460e+10 4.0582e+10 21650
## - smoker    1 1.1811e+11 1.4323e+11 23228
```

```r
summary(step_model)
```

```
##
## Call:
## lm(formula = charges ~ age + bmi + children + smoker + region,
##     data = insurance[-influential_pt, ])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11554.7  -1747.8   -269.4   1567.3  14914.3
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -11054.099    766.923 -14.414  < 2e-16 ***
## age                252.742      9.138  27.658  < 2e-16 ***
## bmi                289.242     22.589  12.804  < 2e-16 ***
## children           518.651    106.362   4.876 1.22e-06 ***
## smokeryes        24795.885    324.361  76.445  < 2e-16 ***
## regionnorthwest   -913.147    366.717  -2.490  0.01290 *
## regionsoutheast  -1054.561    368.845  -2.859  0.00432 **
## regionsouthwest  -1032.353    365.811  -2.822  0.00485 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4496 on 1243 degrees of freedom
## Multiple R-squared:  0.8483, Adjusted R-squared:  0.8475
## F-statistic: 993.1 on 7 and 1243 DF,  p-value: < 2.2e-16
```

Step model did not remove any predictors, so it is the same as the clean model. All predictors are significant at the 0.05 level.

```r
AIC(full_model, clean_model)
```

```
## Warning in AIC.default(full_model, clean_model): models are not all fitted to
## the same number of observations
```

```
##             df      AIC
## full_model  10 27115.51
## clean_model  9 24604.13
```

```r
summary(full_model)$adj.r.squared
```

```
## [1] 0.7494136
```

```r
summary(clean_model)$adj.r.squared
```

```
## [1] 0.8474582
```

```
summary(full_model)$sigma
```

```
## [1] 6062.102
```

```
summary(clean_model)$sigma
```

```
## [1] 4495.626
```

Adjusted R squared: improved by about 10 percentage points, cleaned model explains about 85% of the variation in insurance charges (was 75% before cleaning) RSE: The average error in predicted insurance charges decreased by roughly $1,500, indicating tighter residuals and better prediction accuracy. AIC: A decrease of ~2,683 points is enormous, far greater than the threshold of 10 that indicates a better-fitting model. Influential data points were not random noise, but they were distorting the model fit. The relationships between predictors and charges remain the same, meaning the model's conclusions are robust, not driven by outliers.

Multicollinearity Check

```
X <- model.matrix(clean_model)[, -1]
vif_vals <- diag(solve(cor(X)))
vif_vals
```

```
##             age             bmi        children       smokeryes regionnorthwest
##        1.018914        1.118256        1.005197        1.006465        1.527829
## regionsoutheast regionsouthwest
##        1.666659        1.540579
```

```
cor(select(insurance, age, bmi, children))
```

```
##                age       bmi  children
## age      1.0000000 0.1092719 0.0424690
## bmi      0.1092719 1.0000000 0.0127589
## children 0.0424690 0.0127589 1.0000000
```

All VIF values are well below 5 and all below 2, so the predictors are not correlated enough to inflate the standard errors of the coefficients. On the correlation matrix, all pairwise correlations are very low ($|r| < 0.2$), which means age, BMI, and number of children vary independently.

95% CI and PI for New Case

```
new_person <- data.frame(
  age = 40,
  sex = "female",
  bmi = 25,
  children = 2,
  smoker = "no",
  region = "northwest")

predict(clean_model, newdata = new_person, interval = "confidence", level = 0.95)
```

```
##       fit      lwr      upr
## 1 6410.79 5828.978 6992.602
```

```r
predict(clean_model, newdata = new_person, interval = "prediction", level = 0.95)
```

```
##       fit       lwr      upr
## 1 6410.79 -2428.233 15249.81
```

Predicted Mean: The model predicts an average charge of about $6,389 for a 40-year-old non-smoking female with BMI 25, two children, and living in the northwest region.

95% CI: The CI of $5,803–$6,976 represents the uncertainty in estimating the average medical cost for people with those characteristics. We are 95% confident that the mean insurance charge for individuals with these characteristics lies between $5,800 and $7,000.